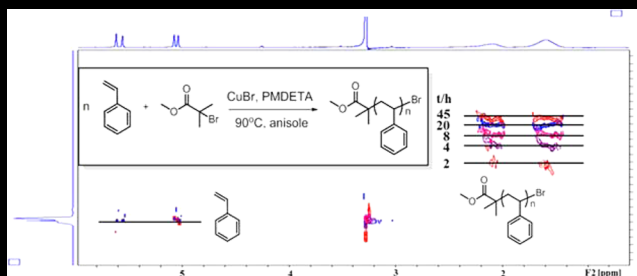


Application of  $^1\text{H}$  DOSY for Facile Measurement of Polymer Molecular Weights

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## ABSTRACT:



The physical properties of polymers are intimately linked to molecular weight (MW). Methods for polymer synthesis are considered “controlled” when they provide polymers with defined MWs and narrow molecular weight distributions (i.e., low polydispersities, PDIs). These key parameters drastically affect the utility of a polymer for a given application. Because synthetic polymers have polydispersity, we speak of their MWs in terms of averages; choice of a relevant average MW is defined by the physical property under consideration. The most common average MWs can be measured via a variety of different techniques such as gel permeation chromatography (GPC), osmometry, static light scattering, matrix-assisted laser desorption-ionization mass spectrometry, viscometry, field-flow fractionation, small-angle X-ray scattering, small-angle neutron scattering, ultracentrifugation, cryoscopy, ebulliometry, and end-group analysis. The weight-average molecular weight is particularly useful to determine certain bulk properties of polymer materials including mechanical properties, thermal properties, and viscoelastic properties. The most common way to determine  $M_w$  is via static light scattering, which relates the intensity of scattered light to molar mass. LS systems are not routinely available in most synthetic laboratories; analysis of  $M_w$  is typically achieved via comparison of GPC retention times to polymer standards.

Although it is easy to obtain molecular weights and molecular weight distributions by GPC analysis, this characterization has many practical drawbacks. For instance, GPC experiments are often time-consuming (typically ~15–60 min per sample); furthermore, GPC instruments often consume large amounts of organic solvent for a single measurement. Also, GPC requires regular calibration to get reliable

information without a LS detector. The expensive columns and detectors are additional burden for GPC maintenance. Additionally, potential solubility problems could rise when changing solvents. Lastly, GPC is incapable of yielding accurate  $M_w$  results at low  $M_w$  range (<2000 g/mol).

In polymer science, 1-D and 2-D NMR techniques have been widely utilized to characterize number-average molecular weight, chemical composition, and microstructures of polymers. Although NMR is very powerful in measuring  $M_n$  via end-group analysis, this method is limited to low molecular weights and situations where polymer backbone resonances do not overlap with end groups. We reasoned that diffusion-ordered NMR spectroscopy (DOSY), which relates the chemical shifts of NMR resonances from a given molecular species to the translational diffusion coefficient of that species, could offer a facile method for accurate determination of molecular weights.

DOSY has been utilized as a powerful tool in recent years to investigate solution state structures and interactions of small organic molecules, organometallic aggregates, and macromolecules. It provides diffusion coefficients of molecules which are related to hydrodynamic radius and molecular weights. Johnson invented and applied DOSY to determine polymer weight distribution. Delsuc reported DOSY-based method to calculate PDIs of linear polymers. DOSY was also utilized by Matyjaszewski to evaluate solvent effects in the atom transfer radical polymerization (ATRP).

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Our lab has recently utilized DOSY to characterize a molecular charm bracelet prepared via ring-closing metathesis and polyrotaxanes prepared via multicomponent ADMET polymerization.

Internal reference systems for  $^1\text{H}$ ,  $^{31}\text{P}$ , and  $^6\text{Li}$  DOSY have been successfully applied to determine the molecular weights of a panel of air/moisture sensitive organometallic complexes and self-assembly systems. Given appropriate internal references, formula weights of organic compounds can be estimated within  $\pm 10\%$  deviation through diffusion coefficient–molecular weight ( $D$ – $M_w$ ) analysis, which is based on the linear correlation of the logarithm of diffusion coefficients ( $\log D$ ) to the logarithm of the molecular weights of the references ( $\log M_w$ ). By evaluating the concentration, viscosity, and density effects on DOSY measurements, accurate molecular weight information has been obtained when the internal reference solution was physically separated from the analyte solution by means of a glass capillary as an insert into the NMR tubes. This was most easily accomplished when the reference and analyte solution concentrations were at or below micromolar level, and therefore the solution properties approached those of the pure solvent.

We envisioned that the principles of the aforementioned reference systems for DOSY could be adopted to create external calibration curves for polymer solutions that would allow one to predict weight-average molecular weights of polymers and monitor the kinetics of living polymerization. For  $^1\text{H}$  DOSY, the acquisition time can be as short as 4 min and the concentration can be at micromolar range, which are ideal for polymer measurements.

This report presents the first use of DOSY technology for the determination of  $M_w$  values for kinetic samples from various living/controlled polymerization. To demonstrate the generality of this approach, we synthesized model polymers of varied composition via three different controlled polymerization methods: atom transfer radical polymerization (ATRP), reversible addition–fragmentation chain transfer (RAFT), and ring-opening metathesis polymerization (ROMP). Kinetic studies were performed on model polymerizations using each of these techniques; the resulting data highlighted the strength of DOSY for convenient monitoring of polymer synthesis.

## ■ RESULTS AND DISCUSSION

For a spherical particle of radius  $R_H$ , the Stokes–Einstein equation describes the diffusion coefficient in eq

$$D = \frac{kT}{6\pi\eta R_H} \quad (1)$$

where  $k$  is the Boltzmann constant,  $T$  is the absolute temperature, and  $\eta$  is the solvent viscosity. On the basis of eq 1, Johnson noted that for a monodispersed polymer system  $D$  can be correlated to molar mass  $M$  described in eq

$$D = AM^\alpha \quad (2)$$

where  $A$  and  $\alpha$  are constants that correlates to different polymers. Equation 2 can be linearized by taking the logarithm of both sides to yield eq

$$\log D = \alpha \log M + \log A \quad (3)$$

For a class of macromolecules with an approximate constancy in density and similarity in shape, the relation of  $D$  and  $M$  is shown in eq

$$M \approx (C/D)^{d_f} \quad (4)$$

where  $d_f$  is the fractal dimension of molecules and  $C$  is a calibration constant. Taking the logarithm of both sides results in eq

$$\log D = -\frac{1}{d_f} \log M + \log C \quad (5)$$

Equation 5 is empirically identical to eq 3 on condition that the shape of a set of analyte compounds remains similar. Moreover, introducing the equation

$$R_H = (3M/4\pi\rho N_A)^{1/3} \quad (6)$$

( $\rho$  is the density of the liquid and  $N_A$  is the Avogadro constant) to the Stokes–Einstein equation (1) results in eq

$$\log D = -\frac{1}{3} \log M + \frac{1}{3} \log \rho - \log \eta - \frac{1}{3} \log \frac{162\pi^2}{k^3 T^3 N_A} \quad (7)$$

To obtain a linear relationship between  $\log D$  and  $\log M$ , convection, density change, and viscosity variation must be well controlled. Since the Stokes–Einstein equation is valid for particles at least 5 times larger than the solvent molecules and under infinitely diluted condition, it is highly applicable to diluted polymer solutions. Diluted solvent will ensure consistent viscosity and density conditions, so that a linear correlation between  $\log D$  and  $\log M$  can be achieved.

We therefore postulated that for all our measurements toward external calibration curves and kinetic analysis of polymerization, highly diluted reaction mixture samples can be measured separately to give reliable and comparable diffusion coefficients by DOSY. By creating the linear calibration curve between  $\log D$  and  $\log M_w$  of known narrowly dispersed polymers, the molecular weight of unknown polymers can be calculated with the measured diffusion coefficients. In order to prove this hypothesis, calibration curves between  $\log D$  and  $\log M_w$  of a set of five selected narrowly dispersed polymers were determined by  $D$ – $M_w$  analysis as a first step.

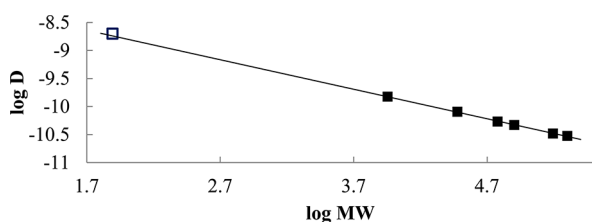
Polystyrene (PS) commercial standards were selected to construct the calibration curve due to its general use as GPC standards. Narrowly dispersed polymers were used to generate the  $D$ – $M_w$  plot calibration to reduce the discrepancy of number-average molecular weight and weight-average molecular weight. Benzene was selected as the solvent due to its relatively high viscosity (0.603 cP at 25 °C) and its ability to solvate a wide variety of organic polymers.

Considering the large molecular weights of polystyrene standards, full signal attenuation during DOSY experiments were achieved (<5%) by prolonged diffusion time to ensure valid values of diffusion coefficients. Table 1 and Figure 1 showed the detailed results of the PS calibration curve in benzene. The high  $r^2$  value of 0.9991 demonstrated excellent linearity of the  $\log D$ – $\log M_w$  curve. Remarkably, if extrapolating the calibration curve from 9000 g/mol to low molecular range, the  $M_w$  of benzene was predicted as 73 g/mol, with only 6.4% deviation of its calculated value. The successful establishment of the polystyrene-based calibration curve in organic system facilitated further kinetic analysis for living polymerization.

**Table 1.**  $D$ – $M_w$  Results of PS Standard Calibration Curve and Predictions

| samples | $M_w(\text{GPC})$<br>(g/mol) | $D$ ( $10^{-11}$ ) | av $M_w$<br>(DOSY)<br>(g/mol) | diff<br>(%) | PDI  |
|---------|------------------------------|--------------------|-------------------------------|-------------|------|
| PS      | 9000                         | $15.02 \pm 0.03$   | 9100                          | 1.1         | 1.06 |
| PS      | 30000                        | $8.08 \pm 0.01$    | 29000                         | −3.3        | 1.06 |
| PS      | 60000                        | $5.40 \pm 0.02$    | 61000                         | 1.7         | 1.04 |
| PS      | 80000                        | $4.69 \pm 0.02$    | 80000                         | 0.0         | 1.07 |
| PS      | 156000                       | $3.31 \pm 0.03$    | 153000                        | −1.9        | 1.06 |
| PS      | 200000                       | $3.00 \pm 0.03$    | 183000                        | −8.5        | 1.06 |
| BEN     | 78                           | 200.0              | 73                            | −6.4        |      |

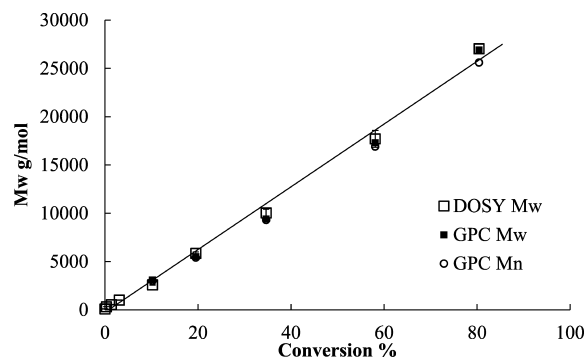
<sup>a</sup><sup>1</sup>H DOSY measurement was performed at 25 °C in benzene-*d*<sub>6</sub>. <sup>b</sup>Equation of the PS standard calibration curve is  $y = -0.537x - 7.697$  ( $R^2 = 0.9991$ ).  $M_w(\text{DOSY})$  was calculated from the calibration curve using the experimental values of  $D$ . <sup>c</sup>Calculated from  $M_w(\text{GPC})$  and  $M_w(\text{DOSY})$ . <sup>d</sup>According to commercial samples. For more details please see

**Figure 1.** PS calibration curve in benzene-*d*<sub>6</sub> for  $M_w$  prediction. Black solid square: PS standards; blue open square: benzene.

We then performed a model ATRP experiment to examine our method. ATRP is the most widely used living radical polymerization because of its high control in polymerization and versatile applications in material science. In ATRP, good control of radical polymerization can be obtained through reversible and repeatable activation/deactivation of dormant species containing propagating radicals in the presence of transition metal complex. In our experiment, ATRP of styrene **1** took place with commonly used copper/nitrogen ligand complex, CuBr, and *N,N,N',N'',N'''*-pentamethyldiethylenetriamine (PMDETA), and alkyl halide, methyl 2-bromoisobutyrate in anisole. Kinetic aliquots were taken from the polymerization solution at each designated time interval.  $M_w$  of each aliquot was determined by both GPC and DOSY. Diffusion coefficients were obtained after full attenuation of NMR signals. Diffusion coefficients of the samples were fitted

to the PS calibration curve (Figure 1) to calculate the  $M_w$ . The aliquot was analyzed by GPC with multiangle laser light scattering in THF eluent to determine the molecular weights of polymers.

The molecular weights of polystyrene **3** linearly increased with conversions showing a typical ATRP behavior (Table 1 and Figure 2). Between the range of 5500–26 900 g/mol (data

**Figure 2.** Conversion– $M_w$  plot of polymerization of styrene by ATRP. Black solid square: DOSY  $M_w$ ; black open square: GPC  $M_w$ ; black open circle: GPC  $M_n$ .

from GPC), the deviation ranged from 0.4% to 6.4%. Most importantly, the results from both GPC and DOSY were closely correlated. The number-average  $M_n$  of the final polymer sample at 2700 min was determined to be 265 000 g/mol by end-group analysis, based on the purified polystyrene sample redissolved in benzene. Thus, the  $M_w$  and  $M_n$  given from NMR data can yield a PDI value of 1.02, consistent with GPC measurement.

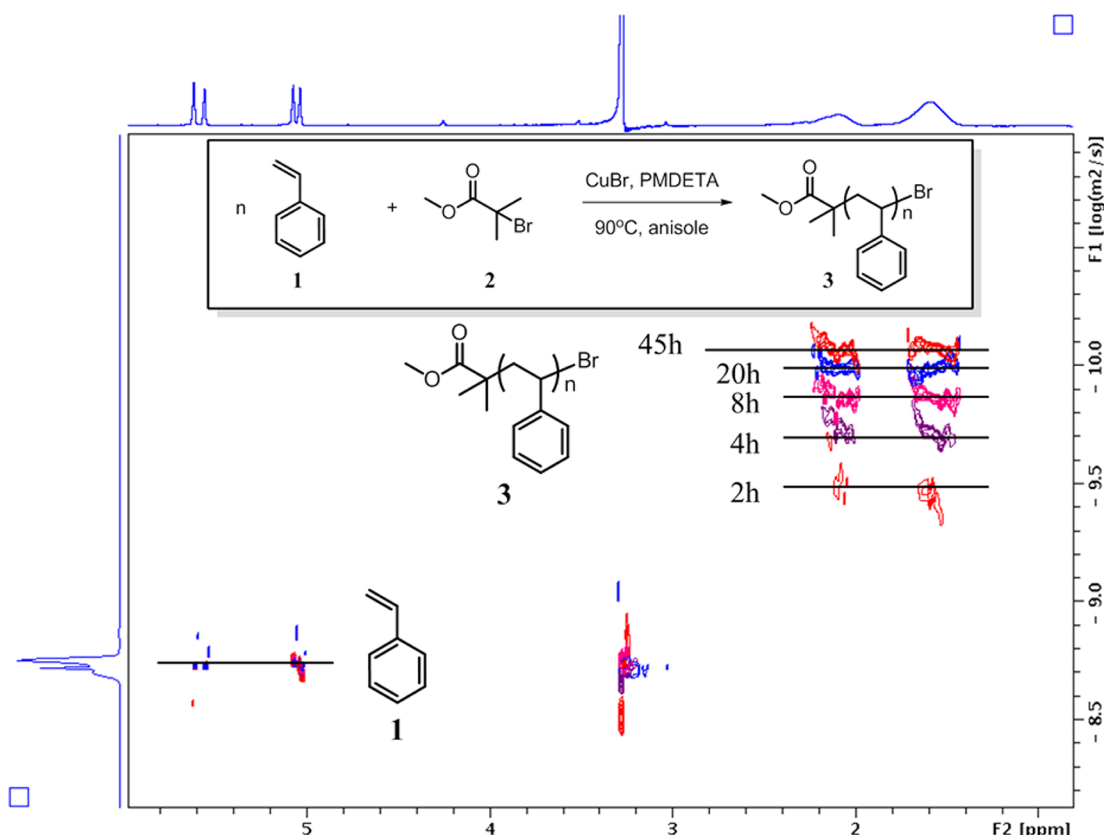
We stacked six DOSY spectra of ATRP aliquots at different time points ranging from 2 h to the end point of 45 h (Figure 3). The diffusion coefficient of styrene monomer **1** remained unchanged at  $17.65 \times 10^{-10}$  m<sup>2</sup>/s, which demonstrated a good viscosity control by solvent dilution. The polystyrene **3** chain continued elongating with increasing values of hydrodynamic radii and reached a  $M_w$  of 27 000 g/mol after 45 h. Since the chemical shifts of monomer and polymer overlapped at aromatic regions, only olefinic and alkyl regions were displayed. The stacked DOSY spectrum was a vivid and informative portrait of this living polymerization process.

This model kinetic study of styrene ATRP demonstrated a close correlation between conventional GPC molecular weight

**Table 2.** ATRP Results from GPC and Externally Referenced DOSY

| $t$ (min) | $M_w(\text{GPC})$ (g/mol) | $D$ ( $10^{-10}$ ) | av $M_w$ (DOSY) (g/mol) | diff (%) | conv (%) | PDI  |
|-----------|---------------------------|--------------------|-------------------------|----------|----------|------|
| 0         |                           | $17.6 \pm 0.5$     | 96                      |          | 0.0      |      |
| 5         |                           | $9.17 \pm 0.02$    | 300                     |          | 0.3      |      |
| 20        |                           | $7.00 \pm 0.09$    | 500                     |          | 1.4      |      |
| 50        |                           | $4.97 \pm 0.03$    | 1000                    |          | 3.1      |      |
| 120       | 3100                      | $3.00 \pm 0.02$    | 2600                    | −16.1    | 10.2     | 1.03 |
| 240       | 5500                      | $1.93 \pm 0.03$    | 5800                    | 5.4      | 19.5     | 1.02 |
| 480       | 9400                      | $1.45 \pm 0.03$    | 10000                   | 6.4      | 34.7     | 1.01 |
| 1200      | 17300                     | $1.07 \pm 0.03$    | 17700                   | 2.3      | 58.1     | 1.02 |
| 2700      | 26900                     | $0.849 \pm 0.002$  | 27000                   | 0.4      | 80.4     | 1.03 |

<sup>a</sup><sup>1</sup>H DOSY measurement was performed at 25 °C in benzene-*d*<sub>6</sub>. <sup>b</sup>Equation of the PS standard calibration curve is  $y = -0.537x - 7.697$  ( $R^2 = 0.9991$ ).  $M_w(\text{DOSY})$  was calculated from the calibration curve using the experimental value of  $D$ . <sup>c</sup>Calculated from  $M_w(\text{GPC})$  and  $M_w(\text{DOSY})$ . <sup>d</sup>Conversion was calculated by <sup>1</sup>H NMR. GPC molecular weight data below 3000 g/mol is not shown due to possible GPC errors at low molecular weight range. For more details please see



**Figure 3.** Stacked DOSY spectra of reaction crude of ATRP. DOSY spectra of 0, 2, 4, 8, 20, and 45 h were stacked. X-axis demonstrates all the  $^1\text{H}$  resonances of the components in solution. Y-axis is the diffusion dimension. Olefinic peaks of styrene were used as internal references to monitor the viscosity change of the dilute crude solution. Because of the heavy overlap of the aromatic regions of styrene and polystyrene, they are not shown in the stacked spectra. Representative ATRP  $^1\text{H}$  DOSY spectrum can be found in

determination method and the DOSY molecular weight determination method over a wide range of molecular weights. Compared to conventional GPC molecular weight measurement, the measuring time for DOSY was 4 min, about one-tenth of the time commonly required for GPC. Moreover, since we used highly diluted samples for  $^1\text{H}$  DOSY measurement, the amount of reaction sample required was less than 10% of that for GPC. As such, this DOSY technique is advantageous to perform large quantity measurement for kinetic analysis as well as smaller reaction setup for polymerization catalyst evaluation.

On the basis of the exciting results from ATRP, we continued to assess reversible addition–fragmentation chain-transfer polymerization (RAFT) kinetic aliquot as another example of living polymerization. The PS calibration curve with the region between 5000 and 30 000 g/mol was enhanced by incorporating four data points obtained from ATRP measurements for more accurate  $M_w$  prediction of RAFT polymers. The enhanced calibration curve ( $y = -0.537x - 7.695$ ) showed an increased  $r^2$  value of 0.9997 with improved accuracy in the lower  $M_w$  range ( ).

We next analyzed a typical RAFT polymerization of styrene **1** in a similar fashion. In our experiment, trithiocarbonate CTA and 2-(dodecylthiocarbonothioylthio)-2-methylpropionic acid (**4**) were used to perform RAFT which was initiated by thermal initiator azobis(isobutyronitrile) (AIBN). We randomly sampled the reaction crude at four time points to verify agreement in molecular weights for both GPC and DOSY experiments (Table ). The molecular weights obtained from GPC and DOSY were within less than 7% deviation between

**Table 3.** RAFT Results from GPC and Externally Referenced DOSY

| $t$<br>(min) | $M_w(\text{GPC})$<br>(g/mol) | $D$<br>( $10^{-10}$ ) | $\text{av } M_w$<br>(DOSY)<br>(g/mol) | diff<br>(%) | conv<br>(%) | PDI  |
|--------------|------------------------------|-----------------------|---------------------------------------|-------------|-------------|------|
| 230          | 2900                         | $2.60 \pm 0.06$       | 3300                                  | 13.8        | 18.7        | 1.04 |
| 590          | 6200                         | $1.80 \pm 0.04$       | 6600                                  | 6.4         | 46.0        | 1.04 |
| 1345         | 9700                         | $1.41 \pm 0.02$       | 10300                                 | 6.2         | 49.4        | 1.06 |
| 1760         | 10600                        | $1.36 \pm 0.03$       | 11000                                 | 3.8         | 53.9        | 1.06 |

<sup>a</sup> $^1\text{H}$  DOSY measurement was performed at 25 °C in benzene- $d_6$ .

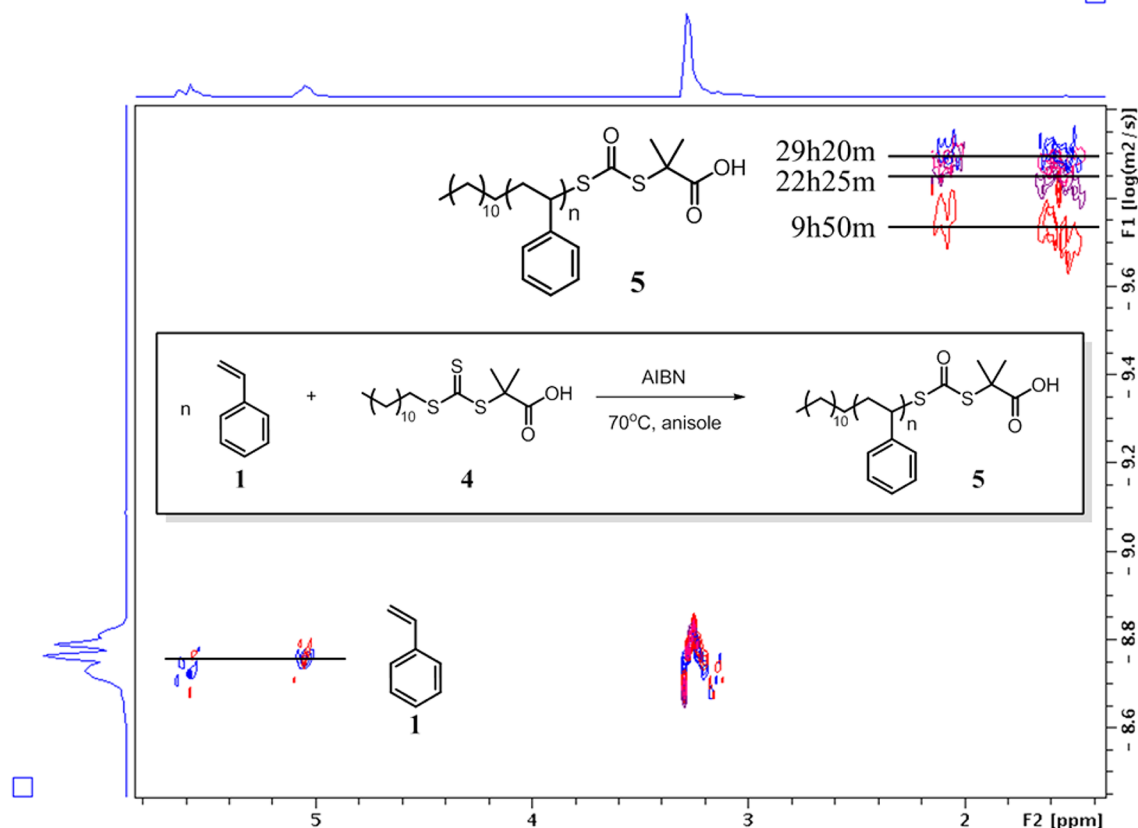
<sup>b</sup>Equation of the PS standard calibration curve is  $y = -0.5369x - 7.6947$  ( $r^2 = 0.9997$ ).  $M_w$  Pred is based on measured  $D$  and the calibration curve. <sup>c</sup>Calculated from  $M_w(\text{GPC})$  and  $M_w(\text{DOSY})$ .

<sup>d</sup>Conversion was calculated by  $^1\text{H}$  NMR. For more details please see

5000 and 90 000 g/mol. In general, higher molecular weight polymer indicated higher agreement between DOSY and GPC values than the lower molecular weight instance. The values of diffusion coefficient of styrene were all close to  $17.65 \times 10^{-10} \text{ m}^2/\text{s}$ , suggesting good viscosity control between each sample. The chain growth of polystyrene **5** over time was conveniently monitored by the decreasing values of diffusion coefficient (Figure ).

Inspired by the results of ATRP and RAFT, we continued to investigate the ring-opening metathesis polymerization (ROMP) to broaden the utility of our method for controlled polymerization.





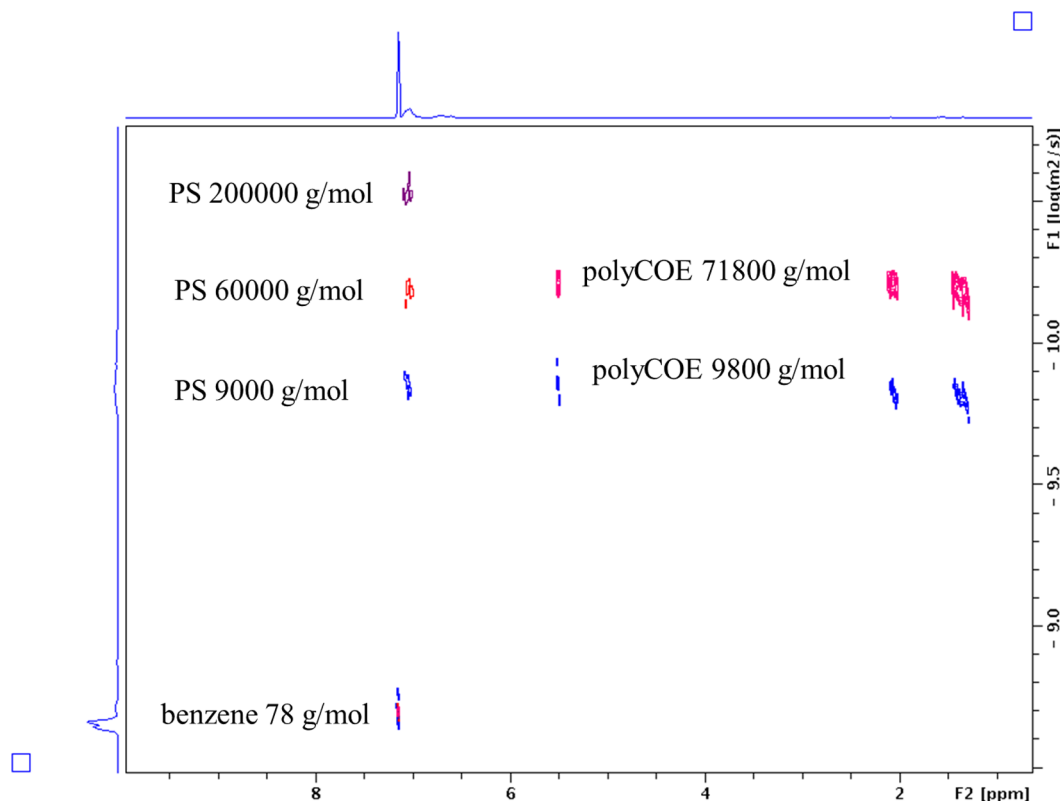
**Figure 4.** Stacked DOSY spectra of reaction crude of RAFT. DOSY spectra of 0 h, 9 h 50 min, 22 h 25 min, and 29 h 20 min were stacked. Olefinic peaks of styrene were used as internal references to monitor the viscosity change of the dilute crude solution. Because of the heavy overlap of the aromatic regions of styrene and polystyrene, they are not shown in the stacked spectra. Representative RAFT  $^1\text{H}$  DOSY spectra can be found in

To begin, two representative polycyclooctenes with  $M_w$  of 9800 and 71 800 g/mol (GPC data) were prepared separately with the second-generation catalyst **8** in dichloromethane. The molecular weights of the polycyclooctene samples were predicted to be 9700 and 68 900 g/mol by the PS calibration curve (Figure ), with 1.2% and 3.9% deviation from the GPC values, respectively. The PS-based calibration plot was tailored again for ROMP type linear polymers by incorporating the two polycyclooctene samples into the existing curve ( ). The  $r^2$  of overall calibration plot increased from 0.9991 to 0.9998, and the molecular weight prediction differences of the polycyclooctene samples were reduced to 0.4% and 3.1% ( ). This serves as an evaluation of whether the calculated  $M_w$  from the new trendline better approximates those of the newly added polymers.

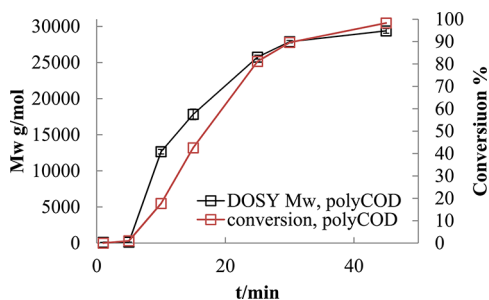
Stacking five DOSY spectra of polystyrene and polycyclooctene (poly(COE)) standards resulted in a spectrum resembling that from a typical gel electrophoresis (Figure ). The benzene and polystyrene standards resonances at  $\delta$  7.15 and 7.00 ppm lined up at various diffusion coefficients according to their molecular weights. The polycyclooctene resonances at  $\delta$  5.50, 2.12, and 1.35 ppm diffused at the same level, with one set of resonances close to polystyrene with a  $M_w$  of 9000 g/mol, and the other set higher than polystyrene with a  $M_w$  of 600 000 g/mol. This intriguing observation implies the possibility of using this method on the determination of molecular weights of biological relevant macromolecules.

The polymerization process of ROMP of 1,5-cyclooctadiene (COD) **6** was initiated by the second-generation catalyst **8** after 5 min (Figure ). The conversion rate was proportional to the increase in molecular weight. Over a period of 45 min, the molecular weight of polyCOD **7** reached 29 600 g/mol with 100% conversion (Figures and ). The  $M_w$ –conversion plot yielded a nonlinear curve with a slope change after 20% conversion. The diffusion coefficients of the newly generated polyCOD **7** resonance at  $\delta$  5.50 ppm decreased from 10 to 45 min from the stacked DOSY spectrum (Figure ).

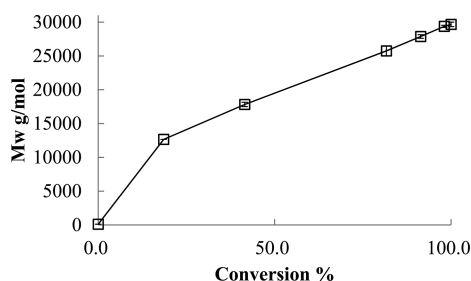
Furthermore, we used the second-generation **8** and third-generation **9** catalysts to polymerize cyclooctene **10** by ROMP in DCM and THF (Figure ). The third-generation catalyst **9** is known to be a fast initiating catalyst for ROMP. The resulting data indicated that **9** initiated the polymerization of COE **10** immediately upon addition to the reaction mixture and completed the reaction in 20 min, generating **11** with a predicted  $M_w$  of 52 700 g/mol. In contrast, the polymerization catalyzed by **8** did not commence until 20 min. However, the final molecular weight of **11** polymerized by **9** was 56 700 g/mol, lower than that catalyzed by **8** (84 900 g/mol). Additionally,  $^1\text{H}$  DOSY and  $^1\text{H}$  NMR experiments showed that ROMP by **8** in THF was slow with low conversion due to the coordinating nature of THF toward active catalytic species in solution. The above experimental results of ROMP demonstrated the practical utility of this method for facile polymerization catalyst evaluation and reaction condition optimization.



**Figure 5.** Stacked DOSY spectra of polystyrene standards and polycyclooctene standards. We chose polystyrene with molecular weight of 9000, 60 000, and 200 000 g/mol and polycyclooctene with molecular weight of 9800 and 71 800 g/mol as representative examples.



**Figure 6.**  $M_w$ - $t$  plot and conversion- $t$  plot of polymerization of 1,5-cyclooctadiene by ROMP. Black open square: weight-average molecular weight provided by  $^1\text{H}$  DOSY. Red open square: conversion calculated from  $^1\text{H}$  NMR.



**Figure 7.**  $M_w$ -conversion plot of polymerization of COD by ROMP under the experimental conditions.

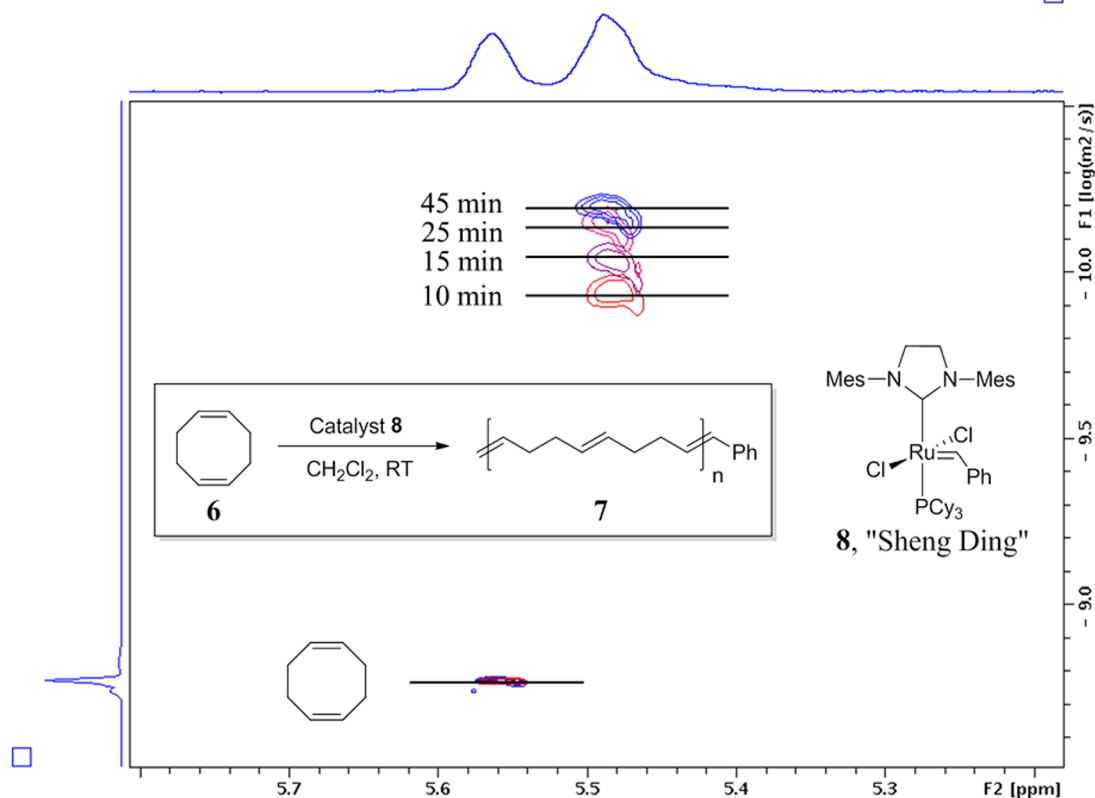
## CONCLUSIONS

A new diffusion NMR-based method to determine the molecular weight of linear and narrowly dispersed polymers is reported. Narrowly dispersed polystyrene-based calibration

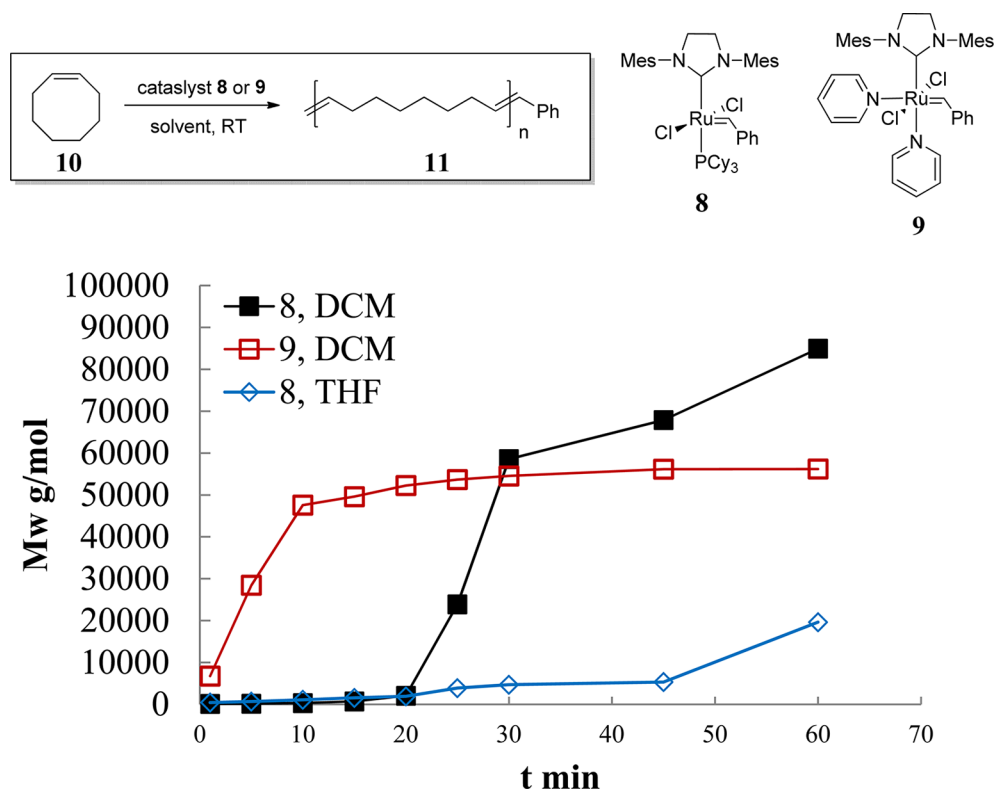
curves have been established using linear relation of  $\log D$  and  $\log M_w$  to determine weight-average molecular weight of linear flexible polymers. The calibration curve can be utilized to accurately monitor the polymerization proceeding of three commonly used polymerizations: ATRP, RAFT, and ROMP. The  $M_w$  of polymers determined by DOSY showed excellent agreement with the corresponding GPC data. The calibration curve of PS was tailored twice with ATRP PS and ROMP polyCOE to adapt to various types of linear polymers for more accurate prediction. Moreover, this method was successfully applied to ROMP to monitor kinetics of polymerization catalyst efficiency. It may replace many conventional polymer molecular weight determination methods including GPC, owing to its short measurement time, low sample loading, and high accuracy at low molecular range. One example of measurement of polydispersity index (defined as  $M_w/M_n$ ) was given in ATRP experiments using DOSY ( $M_w$ ) and end-group analysis ( $M_n$ ) data. More accurate results of PDI can be determined by designing and applying specially modified catalysts or initiators to enhance NMR signals and reduce integration errors.

## EXPERIMENTAL SECTION

**Materials and General Methods.** Benzene- $d_6$  was kept with 4 Å molecular sieves under argon. 1,5-Cyclooctadiene, cyclooctene, Cu(I) Br (99.999%),  $N,N,N',N'',N'''$ -pentamethyldiethylenetriamine (PMDETA), methyl 2-bromoisobutyrate, 2-(dodecylthiocarbonothioylthio)-2-methylpropionic acid (DDMAT), azobis(isobutyronitrile) (AIBN), and styrene were obtained from Aldrich Chemical Co. and used as received. Polystyrene commercial standards were purchased from Polysciences Inc. (9000, 60 000, and 80 000 g/mol), Pressure Chemical Co. (30 000 and 200 000 g/mol), and Shodex Denko America (156 000 g/mol).



**Figure 8.** Stacked DOSY spectra of the reaction crude of ROMP. DOSY spectra of 0, 10, 15, 25, and 45 min were stacked. Olefinic peaks of cyclooctadiene were used as internal references to monitor the viscosity change of the dilute crude solution.



**Figure 9.**  $M_w$ - $t$  plot of polymerization of cyclooctene by ROMP. The monomer-to-catalyst ratio is 5000/1. All reactions were performed in dry dichloromethane or THF at room temperature in 0.5 M concentration.

**ATRP Experiments.** In a Schlenk flask, CuBr (50 mg), PMDETA (60.4 mg), and methyl 2-bromoisobutyrate (63 mg) were placed. The solid reagents containing Schlenk flask was repeatedly vacuumed and backfilled with dry argon for three times. Degassed styrene (7.3 g) and anisole (8 mL) were added to the solid reagent containing Schlenk flask. The first aliquot was taken prior to placing the reaction mixture in a 90 °C oil bath. Kinetic aliquots (0.3 mL) were taken at designated times for DOSY and GPC characterization. For each aliquot, the polymerization was stopped by removing heat followed by exposure of the reaction solution in the air. The polymer solution was filtered through neutral alumina packed column to remove the copper complex.

**RAFT Experiments.** All solid reagents, DDMAT (100 mg), and AIBN (9 mg) were placed in the Schlenk flask, which was repeatedly vacuumed and backfilled with dry argon for three times. Degassed styrene (5.7 g) and anisole (6.3 mL) were added to the Schlenk flask, and then the reaction mixture was degassed by argon bubbling. The reaction mixture was heated at 70 °C with stirring. Aliquots (0.3 mL) were taken at designated times for DOSY and GPC characterization.

**ROMP Experiments.** In a flame-dried and argon-charged 50 mL flask were added 1.696 mg (0.002 mmol) of the second-generation "Sheng Ding" catalyst **8** and 20 mL of dry dichloromethane at room temperature. Timing was started as soon as 1.230 mL (10 mmol) of 1,5-cyclooctadiene was injected into the solution. 1 mL aliquots were extracted via syringe from the reaction mixture at designated times into a scintillation vial with two drops of ethyl vinyl ether. 40  $\mu$ L of the quenched reaction mixture was transferred to an NMR tube and diluted with 1 mL of benzene- $d_6$  for NMR experiments.

**NMR Experiments.** All NMR tubes were flame-dried and evacuated *in vacuo* in advance. NMR experiments were performed at  $25 \pm 1$  °C. All NMR samples were stabilized at 25 °C for 5 min before data collection. For polystyrene standard samples, each NMR tube contained 0.5 mg of polystyrene and 1 mL of benzene- $d_6$ . For ATRP, RAFT, and ROMP samples, each NMR tube contained 40  $\mu$ L of reaction mixture and 1 mL of benzene- $d_6$ . DOSY experiments were performed on a Bruker DRX 300 spectrometer equipped with an Accustar z-axis gradient amplifier and an ATMA BBO probe with a z-axis gradient coil. All experiments were run without spinning to avoid convection. The maximum gradient strength was 0.214 T/m. The standard Bruker pulse program, *stepp1s*, employing a stimulated echo sequence and 1 spoil gradient was utilized. Bipolar rectangular gradients were used with a total duration of 0.5–10 ms (7 ms for polystyrene measurement). Gradient recovery delays were 0.5–1  $\mu$ s. Diffusion times were between 100 and 2000 ms (200 ms for polystyrene measurement). The number of gradient steps was set to be 32. Individual rows of the quasi-2-D diffusion databases were phased and baseline corrected. DOSY spectra were processed by Topspin 1.3 software. Diffusion dimension was generated using *inver Laplace* transform driven by the maximum entropy method. Diffusion coefficients of a chosen narrow chemical shift range were extracted from T1/T2 analysis module of Topspin 1.3 calculated the maximum entropy (GIFA) method developed by Delsuc et al. Unless otherwise mentioned, the DOSY spectrum was processed without additional change of processing parameters.

**GPC Characterization.** GPC analysis of polymer was performed by two PLgel 10  $\mu$ m mixed-B LS columns (Polymer Laboratories) connected in series with a DAWN EOS multiangle laser light scattering (MALLS) detector (Wyatt Technology), an Optilab DSP differential refractometer (Wyatt Technology), and a Viscostar viscometer (Wyatt Technology) in THF eluent. The analysis was performed with a flow rate of 1 mL/min at room temperature. Absolute molecular weights and PDI were calculated from  $dn/dc$  values that were obtained by assuming 100% mass elution from the GPC columns.

## ■ ASSOCIATED CONTENT

### ■ Supporting Information

NMR spectra and experimental procedures. This material is available free of charge via the Internet at

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### Notes

The authors declare no competing financial interest.

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## ■ REFERENCES

- (1) Mark, J.; Ngai, K.; Graessley, W.; Mandelkern, L.; Samulski, E.; Koenig, J.; Wignall, G. *Physical Properties of Polymers*, 3rd ed.; Cambridge University Press: Cambridge, 2004.
- (2) Flory, P. J. *Principles of Polymer Chemistry*; Cornell University Press: Ithaca, NY, 1953.
- (3) Allcock, H. R.; Lampe, F. W.; Mark, J. E. In *Contemporary Polymer Chemistry*, 3rd ed.; Pearson/Prentice Hall: Upper Saddle River, NJ, 2003; pp 393–491.
- (4) Stevens, M. P. In *Polymer Chemistry: An Introduction*, 3rd ed.; Oxford University Press: New York, 1999; pp 35–60.
- (5) Young, R. J.; Lovell, P. A. In *Introduction to Polymers*, 3rd ed.; CRC Press: Boca Raton, FL, 2011; pp 269–341.
- (6) Carraher, C. E. J. In *Carraher's Polymer Chemistry*, 8th ed.; CRC Press: Boca Raton, FL, 2011; pp 55–91.
- (7) Fried, J. R. In *Polymer Science and Technology*, 1st ed.; Prentice Hall PTR: Englewood Cliffs, NJ, 1995; pp 110–131.
- (8) Cowie, J. M. G.; Arrighi, V. In *Polymers: Chemistry and Physics of Modern Materials*, 3rd ed.; CRC Press: Boca Raton, FL, 2008; pp 229–252.
- (9) Young, R. J.; Lovell, P. A. In *Introduction to Polymers*, 3rd ed.; CRC Press: Boca Raton, FL, 2011; pp 343–380.
- (10) Elias, H.-G. In *Macromolecules*; Wiley-VCH: Darmstadt, 2005; Vol. 1, pp 109–151.
- (11) Cohen, Y.; Avram, L.; Frish, L. *Angew. Chem., Int. Ed.* **2005**, *44*, 520–554.
- (12) Johnson, C. S., Jr. *Prog. Nucl. Magn. Reson. Spectrosc.* **1999**, *34*, 203–256.
- (13) Morris, K. F.; Johnson, C. S. *J. Am. Chem. Soc.* **1992**, *114*, 3139–3141.
- (14) Chen, A.; Wu, D.; Johnson, C. S. *J. Am. Chem. Soc.* **1995**, *117*, 7965–7970.
- (15) Morris, K. F.; Johnson, C. S. *J. Am. Chem. Soc.* **1993**, *115*, 4291–4299.
- (16) Valentini, M.; Vaccaro, A.; Rehor, A.; Napoli, A.; Hubbell, J. A.; Tirelli, N. *J. Am. Chem. Soc.* **2004**, *126*, 2142–2147.
- (17) Cain, J. B.; Zhang, K.; Betts, D. E.; DeSimone, J. M.; Johnson, C. S. *J. Am. Chem. Soc.* **1998**, *120*, 9390–9391.
- (18) Masuda, A.; Ushida, K.; Koshino, H.; Yamashita, K.; Kluge, T. *J. Am. Chem. Soc.* **2001**, *123*, 11468–11471.
- (19) Pregosin, P. S.; Kumar, P. G. A.; Fernández, I. *Chem. Rev.* **2005**, *105*, 2977–2998.
- (20) Pregosin, P. S.; Martinez-Viviente, E.; Kumar, P. G. A. *Dalton Trans.* **2003**, 4007–4014.
- (21) Dehner, A.; Kessler, H. *ChemBioChem* **2005**, *6*, 1550–1565.
- (22) Li, D.; Keresztes, I.; Hopson, R.; Williard, P. G. *Acc. Chem. Res.* **2009**, *42*, 270–280.
- (23) Viéville, J.; Tanty, M.; Delsuc, M.-A. *J. Magn. Reson.* **2011**, *212*, 169–173.



- (24) Braunecker, W. A.; Tsarevsky, N. V.; Gennaro, A.; Matyjaszewski, K. *Macromolecules* **2009**, *42*, 6348–6360.
- (25) Clark, P. G.; Guidry, E. N.; Chan, W. Y.; Steinmetz, W. E.; Grubbs, R. H. *J. Am. Chem. Soc.* **2010**, *132*, 3405–3412.
- (26) Momcilovic, N.; Clark, P. G.; Boydston, A. J.; Grubbs, R. H. *J. Am. Chem. Soc.* **2011**, *133*, 19087–19089.
- (27) Ma, L.; Hopson, R.; Li, D.; Zhang, Y.; Williard, P. G. *Organometallics* **2007**, *26*, 5834–5839.
- (28) Li, D.; Hopson, R.; Li, W.; Liu, J.; Williard, P. G. *Org. Lett.* **2008**, *10*, 909–911.
- (29) Li, D.; Sun, C.; Liu, J.; Hopson, R.; Li, W.; Williard, P. G. *J. Org. Chem.* **2008**, *73*, 2373–2381.
- (30) Liu, J.; Li, D.; Sun, C.; Williard, P. G. *J. Org. Chem.* **2008**, *73*, 4045–4052.
- (31) Kagan, G.; Li, W.; Hopson, R.; Williard, P. G. *Org. Lett.* **2009**, *11*, 4818–4821.
- (32) Li, W.; Kagan, G.; Yang, H.; Cai, C.; Hopson, R.; Dai, W.; Sweigart, D. A.; Williard, P. G. *Organometallics* **2010**, *29*, 1309–1311.
- (33) Li, W.; Kagan, G.; Yang, H.; Cai, C.; Hopson, R.; Sweigart, D. A.; Williard, P. G. *Org. Lett.* **2010**, *12*, 2698–2701.
- (34) Kagan, G.; Li, W.; Li, D.; Hopson, R.; Williard, P. G. *J. Am. Chem. Soc.* **2011**, *133*, 6596–6602.
- (35) Kagan, G.; Li, W.; Sun, C.; Hopson, R.; Williard, P. G. *J. Org. Chem.* **2011**, *76*, 65–70.
- (36) Matyjaszewski, K. *Macromolecules* **2012**, *45*, 4015–4039.
- (37) Braunecker, W. A.; Matyjaszewski, K. *Prog. Polym. Sci.* **2007**, *32*, 93–146.
- (38) Matyjaszewski, K.; Xia, J. H. *Chem. Rev.* **2001**, *101*, 2921–2990.
- (39) Moad, G.; Rizzardo, E.; Thang, S. H. *Aust. J. Chem.* **2009**, *62*, 1402–1472.
- (40) Moad, G.; Rizzardo, E.; Thang, S. H. *Aust. J. Chem.* **2006**, *59*, 669–692.
- (41) Moad, G.; Rizzardo, E.; Thang, S. H. *Aust. J. Chem.* **2005**, *58*, 379–410.
- (42) Moad, G.; Rizzardo, E.; Thang, S. H. *Polymer* **2008**, *49*, 1079–1131.
- (43) Bielawski, C. W.; Grubbs, R. H. *Prog. Polym. Sci.* **2007**, *32*, 1–29.
- (44) Vougioukalakis, G. C.; Grubbs, R. H. *Chem. Rev.* **2010**, *110*, 1746–1787.
- (45) Sutthasupa, S.; Shiotsuki, M.; Sanda, F. *Polym. J.* **2010**, *42*, 905–915.
- (46) Leitgeb, A.; Wappel, J.; Slugovc, C. *Polymer* **2010**, *51*, 2927–2946.
- (47) Boyer, C.; Bulmus, V.; Davis, T. P.; Ladmiral, V.; Liu, J. Q.; Perrier, S. *Chem. Rev.* **2009**, *109*, 5402–5436.
- (48) Floquet, S. b.; Brun, S. b.; Lemonnier, J.-F. o.; Henry, M.; Delsuc, M.-A.; Prigent, Y.; Cadot, E.; Taulelle, F. *J. Am. Chem. Soc.* **2009**, *131*, 17254–17259.
- (49) Reynolds, W. W.; Gebhart, H. J. *J. Chem. Eng. Data* **1960**, *5*, 220–225.
- (50) Stark, A.; Ajam, M.; Green, M.; Raubenheimer, H. G.; Ranwell, A.; Ondruschka, B. *Adv. Synth. Catal.* **2006**, *348*, 1934–1941.
- (51) Delsuc, M. A.; Malliavin, T. E. *Anal. Chem.* **1998**, *70*, 2146–2148.